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Slot: B11 + B12

Experiments

Experiment 1

1. Write a Python program to open a file named student.txt in read mode. Handle the exception if the file does not exist.

Code:

```
p1.py
1  try:
2      file = open("student.txt", "r")
3      content = file.read()
4      print(content)
5      file.close()
6  except FileNotFoundError:
7      print("Error: student.txt does not exist.")
```

Output:

Terminal

```
Error: student.txt does not exist.
```

Experiment 2

2. Write a Python program using try-except to read a file and display an appropriate message when an I/O error occurs.

Code:

```
p2.py
1  try:
2      file = open("data.txt", "r")
3      content = file.read()
4      print(content)
5      file.close()
6  except IOError:
7      print("I/O Error: Unable to read the file.")
```

Output:

Terminal

```
I/O Error: Unable to read the file.
```

Experiment 3

3. Write a Python program to create a directory named College using the os module.

Code:

```
p3.py
1  import os
2
3  if not os.path.exists("College"):
4      os.mkdir("College")
5      print("College directory created.")
6  else:
7      print("College directory already exists.")
```

Output:

Terminal

College directory created.

Experiment 4

4. Write a Python program to display the current working directory using `os.getcwd()`.

Code:

```
p4.py
1  import os
2
3  current_directory = os.getcwd()
4
5  print("Current working directory:")
6  print(current_directory)
```

Output:

Terminal

Current working directory:
C:\Users\Student\PythonAssignment

Experiment 5

5. Write a Python program to create a directory named OldFolder and rename it to NewFolder.

Code:

p5.py

```
1  import os
2
3  os.mkdir("OldFolder")
4  print("OldFolder created.")
5
6  os.rename("OldFolder", "NewFolder")
7  print("OldFolder renamed to NewFolder.")
```

Output:

Terminal

```
OldFolder created.
OldFolder renamed to NewFolder.
```

Experiment 6

6. Write a Python program to delete an empty directory named TempFolder using `os.rmdir()`.

Code:

p6.py

```
1  import os
2
3  if os.path.exists("TempFolder"):
4      os.rmdir("TempFolder")
5      print("TempFolder deleted.")
6  else:
7      print("TempFolder does not exist.")
```

Output:

Terminal

```
TempFolder deleted.
```

Experiment 7

7. Write a Python program that checks whether a file exists before attempting to open it. Handle the exception appropriately.

Code:

p7.py

```
1  import os
2
3  try:
4      if os.path.exists("student.txt"):
5          file = open("student.txt", "r")
6          print(file.read())
7          file.close()
8      else:
9          raise FileNotFoundError
10 except FileNotFoundError:
11     print("Error: File does not exist.")
```

Output:

Terminal

Error: File does not exist.

Experiment 8

8. Write a Python program using try-except-finally to open and read data.txt. Display a message in the finally block after the file operation is completed.

Code:

p8.py

```
1  try:
2      file = open("data.txt", "r")
3      print(file.read())
4      file.close()
5  except FileNotFoundError:
6      print("Error: data.txt not found.")
7  finally:
8      print("File operation completed.")
```

Output:

Terminal

Error: data.txt not found.
File operation completed.

Experiment 9

9. Write a Python program to read all the contents of input.txt, convert the contents into a string, and write them into output.txt.

Code:

p9.py

```
1  with open("input.txt", "r") as file:
2      data = file.read()
3
4  text = str(data)
5
6  with open("output.txt", "w") as file:
7      file.write(text)
8
9  print("Contents copied to output.txt.")
```

Output:

Terminal

Contents copied to output.txt.

Experiment 10

10. Write a Python program to read input.txt, count the occurrence of each word, and store the word-frequency results in output.txt.

Code:

p10.py

```
1  with open("input.txt", "r") as file:
2      text = file.read().lower()
3
4  words = text.split()
5  frequency = {}
6
7  for word in words:
8      word = word.strip(".,!?")
9      frequency[word] = frequency.get(word, 0) + 1
10
11 with open("output.txt", "w") as file:
12     for word, count in frequency.items():
13         file.write(word + " : " + str(count) + "\n")
14
15 print("Word frequencies stored in output.txt.")
```

Output:

Terminal

Word frequencies stored in output.txt.

Experiment 11

11. Write a Python program to read a text file and count the number of lines, words, and characters in the file.

Code:

```
p11.py
1  with open("input.txt", "r") as file:
2      text = file.read()
3
4  lines = text.splitlines()
5  words = text.split()
6  characters = len(text)
7
8  print("Number of lines:", len(lines))
9  print("Number of words:", len(words))
10 print("Number of characters:", characters)
```

Output:

Terminal

```
Number of lines: 3
Number of words: 12
Number of characters: 72
```

Experiment 12

12. Write a Python program to copy the contents of input.txt into backup.txt. Handle the situation where input.txt does not exist.

Code:

```
p12.py
1  try:
2      with open("input.txt", "r") as source:
3          data = source.read()
4
5      with open("backup.txt", "w") as backup:
6          backup.write(data)
7
8      print("File copied to backup.txt.")
9  except FileNotFoundError:
10     print("Error: input.txt does not exist.")
```

Output:

Terminal

```
File copied to backup.txt.
```

Experiment 13

13. Write a Python program that asks the user for a filename, attempts to open it, and displays a suitable message if the file cannot be opened.

Code:

```
p13.py
1  filename = input("Enter filename: ")
2
3  try:
4      file = open(filename, "r")
5      print("File opened successfully.")
6      print(file.read())
7      file.close()
8  except FileNotFoundError:
9      print("Error: File not found.")
10 except IOError:
11     print("Error: File cannot be opened.")
```

Output:

Terminal

```
Enter filename: student.txt
Error: File not found.
```

Experiment 14

14. Write a Python program to create a directory, create a text file inside it, write some content into the file, and then display the current working directory.

Code:

```
p14.py
1  import os
2
3  folder = "College"
4  os.makedirs(folder, exist_ok=True)
5
6  file_path = os.path.join(folder, "notes.txt")
7
8  with open(file_path, "w") as file:
9      file.write("This file is inside the College directory.")
10
11 print("Directory and file created.")
12 print("Current working directory:")
13 print(os.getcwd())
```

Output:

Terminal

```
Directory and file created.  
Current working directory:  
C:\Users\Student\PythonAssignment
```

Experiment 15

15. Write a Python program using try, except, and finally to safely perform a file-reading operation and ensure that a completion message is displayed whether an exception occurs or not.

Code:

```
p15.py  
  
1  try:  
2      file = open("sample.txt", "r")  
3      content = file.read()  
4      print("File contents:")  
5      print(content)  
6      file.close()  
7  except FileNotFoundError:  
8      print("Error: sample.txt not found.")  
9  except IOError:  
10     print("Error: Unable to read the file.")  
11  finally:  
12     print("File reading operation completed.")
```

Output:

Terminal

```
Error: sample.txt not found.  
File reading operation completed.
```